A Report

ON

Project Based Learning: Digital Electronics

“RFID Door Lock using Arduino”

BY

Paresh Sonnekar - 1032231999

Anish Tilekar - 1032231904

Pranav Raikar - 1032240060

Akshwaj Rao - 1262240046
Aaditya Sharma -1032231950

TY B.Tech Computer Science And Engineering

Under the Guidance of:

Subject Teacher: Prof. Vinod Kumar

MIT-WORLD PEACE UNIVERSITY

School of Computer Science and Engineering

Pune – 411038

CERTIFICATE

This is to certify that the Project-Based Learning (PBL) Report entitled

“RFID Door Lock using Arduino”

has been successfully completed and submitted by

- **Paresh Sonnekar (1032231999)**
- **Anish Tilekar (1032231904)**
- **Pranav Raikar (1032240060)**
- **Akshwaj Rao (1262240046)**
- **Aaditya Sharma (1032231950)**

of TY B.Tech (Computer Science and Engineering) in partial fulfillment of the requirements for the course Project Based Learning: Digital Electronics during the Academic Year 2025–2026.

This project report has been approved as a record of authentic work carried out under the guidance of the undersigned.

Project Guide / Subject Teacher:
(Signature)
Name: ________________________

Department of Computer Science and Engineering
MIT-WPU, Pune

Head of Department:
(Signature)
Name: ________________________

Date: ____________________

Place: Pune

Abstract

The RFID Door Lock system provides a secure, convenient, and cost-effective solution for controlling door access using RFID technology. The project replaces traditional mechanical keys with RFID cards or tags, offering faster authentication and improved reliability. This system operates by detecting authorized RFID tags and triggering a servo motor to unlock or lock the door accordingly. The goal is to enhance security and reduce human effort while maintaining simplicity in design and operation. The project is designed to demonstrate practical IoT integration through microcontrollers, sensors, and real-time communication.

Index

Introduction:

The RFID Door Lock project aims to provide a convenient and secure way to lock and unlock doors using RFID technology. This system replaces traditional keys with RFID tags, allowing users to unlock doors using proximity. The project involves two main components: a bolt latch (lock) and a RFID sensing module. The system is designed to be user friendly with hassle free installation and features like fail safes and overrides to ensure access even if keys are lost. Overall, the RFID door lock project focuses on enhancing security and convenience for consumers by offering a simple and cost-effective upgrade to traditional door locks.

Problem Statement

Traditional door locks using keys or number combinations are vulnerable to theft, duplication, or misuse. The need for a modern, automated, and secure locking mechanism has led to the use of RFID-based access systems.

Objective

- To design and implement an RFID-based door locking system using Arduino.

- To ensure secure access using unique RFID tags.

- To demonstrate IoT integration in real-life security systems.

Scope and Limitations

Scope:

- Can be applied in homes, offices, hotels, and institutions.
- Easy to modify for additional features like mobile control or database storage.

Limitations:

- Limited memory for storing authorized tag IDs.
- Requires continuous power supply.

Literature Review / Existing Systems

(a) Related Research

Several studies have been conducted on RFID-based access control systems using microcontrollers such as Arduino. Most research focuses on developing low-cost, reliable, and fast security systems for homes, offices, and institutions. RFID technology is widely preferred due to its contactless operation, quick response, and ease of user identification. Typically, RFID readers like EM-18 (125 kHz) and RC522 (13.56 MHz) are used to read unique tag IDs, which are then verified by the microcontroller.

Many researchers have implemented automatic door locking using servo motors or electromagnetic locks, enabling smooth mechanical operation. Recent advancements include IoT integration, where WiFi modules are used to send entry logs to cloud platforms for remote monitoring and alerts. Studies also discuss security concerns such as RFID tag cloning and unauthorized access, and suggest improvements like encrypted communication, secure smart cards, and multi-factor authentication for better protection.

(b) Comparison of Existing Solutions

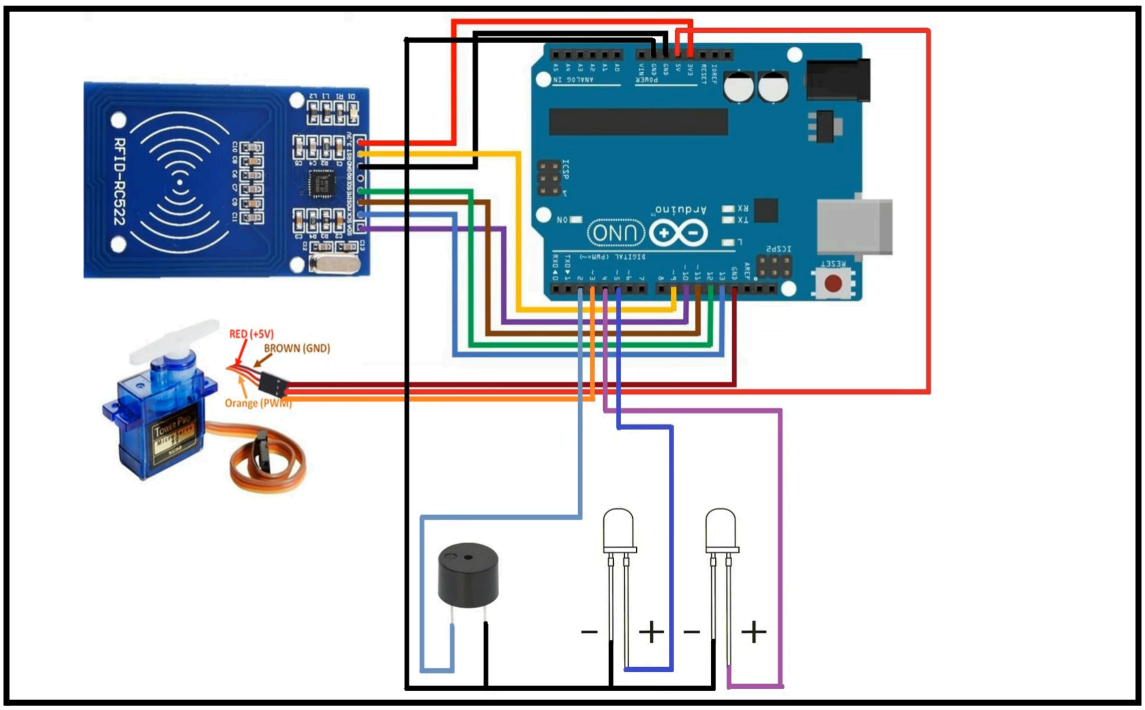
Various door access control systems exist today, including traditional key-based locks, keypad-based systems, biometric systems, and RFID-based systems. Traditional lock systems are simple and inexpensive but suffer from key loss, duplication, and lack of monitoring. Keypad-based systems improve security slightly but are vulnerable to password sharing and shoulder surfing. Biometric systems provide high security and user identity verification, but they are expensive and sensitive to environmental conditions.

RFID-based door locking systems offer a good balance between security, cost, and ease of use. They provide contactless access, fast authentication, and easy scalability for adding or removing users. Compared to traditional and keypad systems, RFID systems are more secure and convenient, while being more affordable than biometric systems. When combined with IoT, RFID systems further enable real-time monitoring and remote access control, making them suitable for modern smart security applications.

The architecture of the RFID-based door locking system follows a centralized embedded structure with the Arduino Uno as the main controller. The RFID reader scans the card and sends the unique ID to the Arduino for verification. If the ID is valid, the Arduino activates the servo motor to unlock the door and provides feedback through a buzzer or display. In case of an invalid card, access is denied and an alert is generated. A regulated power supply ensures stable operation, and an optional WiFi module enables IoT-based remote monitoring. This architecture provides secure, fast, and reliable access control.

System Design

Block diagram



b. Architecture of the system

The architecture of the RFID-based door locking system is designed using a centralized embedded control model, where the Arduino Uno acts as the core controller of the entire system. All input and output devices are directly interfaced with the Arduino. The RFID reader serves as the primary input unit, which scans the RFID tag and sends the unique tag ID to the Arduino through serial/SPI communication. The Arduino processes this data by comparing it with the list of authorized IDs stored in its memory. Based on the verification result, control signals are generated to operate the locking mechanism.

The output section of the architecture consists of a servo motor, buzzer, and optional display unit. When a valid RFID card is detected, the Arduino sends a PWM signal to the servo motor to unlock the door, while the buzzer provides audio feedback. If an invalid card is scanned, access is denied and the buzzer alerts the user. The system is powered using a regulated power supply to ensure stable operation. If IoT integration is included, a WiFi module is added to the architecture to transmit access data to a cloud platform for remote monitoring and logging. This architecture ensures fast response, secure authentication, and reliable real-time operation.

c. Hardware requirements

- Arduino Uno R3
- RFID Reader & Tags
- Servo Motor
- Bolt Latch
- Breadboard
- LEDs
- Buzzer
- Foam Board
- Jumper Wires

d. Software requirements

- Arduino IDE – For writing, compiling, and uploading the program to the Arduino Uno
- USB Driver for Arduino Uno – For communication between the computer and Arduino board
- RFID Library (MFRC522) – To interface the RC522 RFID reader with Arduino
- SPI Library – Required for communication between Arduino and RFID module
- Servo Library – To control the servo motor for door locking and unlocking
- LiquidCrystal / LiquidCrystal_I2C Library – (Optional) For displaying messages on 16×2 LCD
- ESP8266/ESP32 WiFi Library – (Optional, if IoT is used) For internet connectivity
- MQTT / HTTP Library – (Optional) For cloud data transmission and monitoring

Methodology

Working Principle

The working principle of the RFID-based door locking system is based on contactless identification and automatic access control. When a user brings an RFID card near the RFID reader, the reader detects the card using radio frequency signals and reads the unique identification number (UID) stored in it. This UID is then sent to the Arduino Uno for verification. The Arduino compares the received UID with the pre-stored authorized card numbers.

If the scanned UID matches with any authorized UID, the Arduino sends a control signal to the servo motor to unlock the door, and a buzzer or display provides a confirmation message as “Access Granted.” If the UID does not match, the system denies access, keeps the door locked, and activates the buzzer as an alert. After a short delay, the door automatically locks again, making the system suitable for secure, fast, and automatic access control.

b. Algorithm

1. Start the system.
2. Initialize the Arduino, RFID reader, servo motor, buzzer, and LCD (if used).
3. Display the message “Scan Your Card”.
4. Wait for an RFID card to be scanned.
5. Read the unique ID (UID) of the scanned RFID card.

6. Compare the scanned UID with the stored authorized UID list.
7. If the UID matches:

   Display “Access Granted”.

   Activate the servo motor to unlock the door.

   Turn ON the buzzer briefly for confirmation.

   Wait for a few seconds.

   Lock the door again by resetting the servo motor position.

8. If the UID does NOT match:

   Display “Access Denied”.

   Activate the buzzer for an alert sound.

9. Repeat the process from Step 3 for the next scan.

10. End.

Implementation

a. Hardware Setup

The hardware setup consists of the Arduino Uno as the main microcontroller, an RFID reader (RC522) to scan RFID cards, a servo motor to control the lock mechanism, a buzzer for alert indication, and an LCD for displaying system status. A Wi-Fi module such as the ESP8266 or ESP32 is used for IoT connectivity if remote monitoring is required. All components are powered through the Arduino and connected using jumper wires on a breadboard or PCB as per the circuit design. The correct interfacing of sensors, actuators, and power supply ensures smooth operation of the system.

b. Software Coding and Integration

The software is developed using the Arduino IDE in Embedded C. Libraries such as MFRC522 (for RFID), Servo (for motor control), and SPI (for communication with the RFID module) are included in the code. The program logic handles card detection, UID verification, motor control for locking and unlocking, and buzzer or display output. If IoT is implemented, Wi-Fi and MQTT/HTTP libraries are added to transmit access data to a cloud platform. All modules are integrated into a single program to ensure coordinated functioning.

c. Communication Protocols (Wi-Fi, MQTT, etc.)

Wi-Fi communication is used to connect the system to the internet for real-time data monitoring and control. MQTT protocol is commonly used for lightweight message transmission between the Arduino system and the cloud server due to its low bandwidth usage and fast delivery. The system publishes access status such as valid or invalid card detection to an MQTT broker, enabling remote monitoring through a web or mobile application. This makes the system suitable for smart security and IoT-based automation.

Results & Discussion

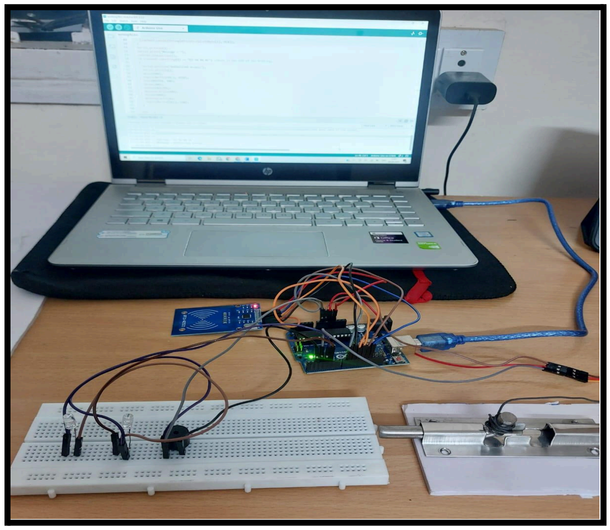
```
#include <SPI.h>
#include <MFRC522.h>

#define RST_PIN 9
#define SS_PIN  10

MFRC522 mfrc522(SS_PIN, RST_PIN);

void setup() {
  Serial.begin(9600);// Initialize serial communications with the PC
  while (!Serial);// Do nothing if no serial port is opened (added for Arduinos based on ATMEGA32U4)
  SPI.begin();// Init SPI bus
  mfrc522.PCD_Init();// Init MFRC522
  delay(4);// Optional delay. Some board do need more time after init to be ready, see Readme
  mfrc522.PCD_DumpVersionToSerial();// Show details of PCD - MFRC522 Card Reader details
  Serial.println(F("Scan PICC to see UID, SAK, type, and data blocks..."));
}

void loop() {
  // Reset the loop if no new card present on the sensor/reader. This saves the entire process when idle.
  if ( ! mfrc522.PICC_IsNewCardPresent()) {
    return;
  }
// Select one of the cards
  if ( ! mfrc522.PICC_ReadCardSerial()) {
    return;
  }
// Dump debug info about the card; PICC_HaltA() is automatically called
  mfrc522.PICC_DumpToSerial(&(mfrc522.uid));
}
```

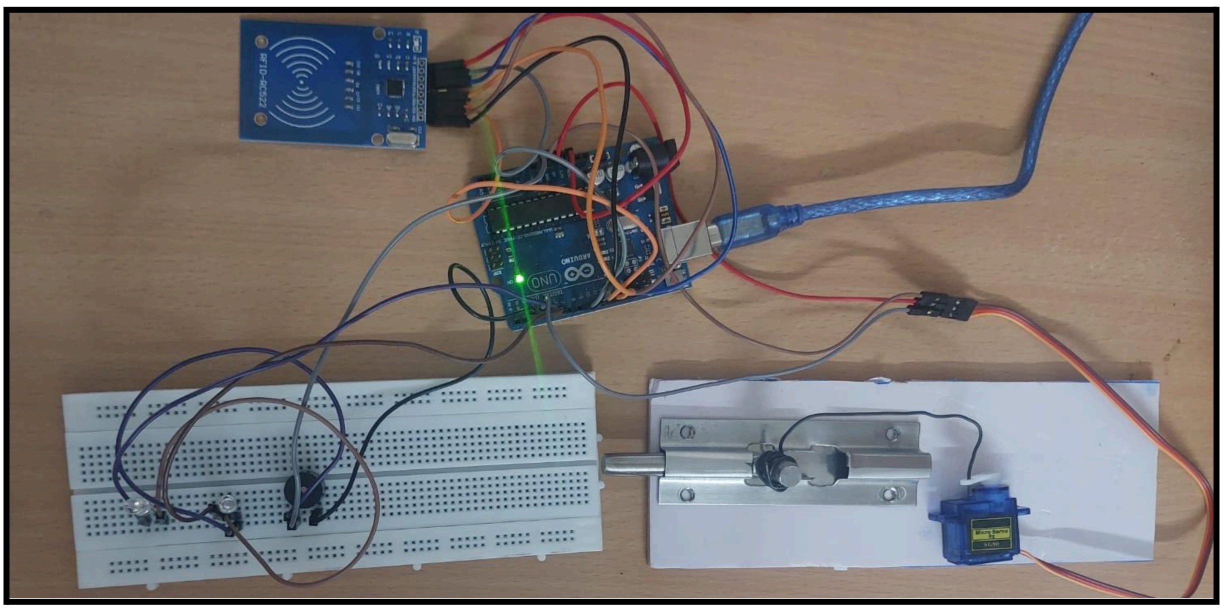
➢ **Code to control the working of Door Lock System:**

```
#include <SPI.h>
#include <MFRC522.h>
#include <Servo.h>

#define SS_PIN 10
#define RST_PIN 9
#define LED_G 5 //define green LED pin
#define LED_R 4 //define red LED
#define BUZZER 2 //buzzer pin
MFRC522 mfrc522(SS_PIN, RST_PIN);   // Create MFRC522 instance.
Servo myServo; //define servo name

void setup()
{
  Serial.begin(9600);   // Initiate a serial communication
  SPI.begin();      // Initiate  SPI bus
  mfrc522.PCD_Init();   // Initiate MFRC522
  myServo.attach(3); //servo pin
  myServo.write(0); //servo start position
  pinMode(LED_G, OUTPUT);
  pinMode(LED_R, OUTPUT);
  pinMode(BUZZER, OUTPUT);
  noTone(BUZZER);
  Serial.println("Put your card to the reader...");
  Serial.println();

}
void loop()
{
  // Look for new cards
```

```
{
  // Look for new cards
  if ( ! mfrc522.PICC_IsNewCardPresent())
  {
    return;
  }
  // Select one of the cards
  if ( ! mfrc522.PICC_ReadCardSerial())
  {
    return;
  }
  //Show UID on serial monitor
  Serial.print("UID tag :");
  String content= "";
  byte letter;
  for (byte i = 0; i < mfrc522.uid.size; i++)
  {
     Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
     Serial.print(mfrc522.uid.uidByte[i], HEX);
     content.concat(String(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " "));
     content.concat(String(mfrc522.uid.uidByte[i], HEX));
  }
  Serial.println();
  Serial.print("Message : ");
  content.toUpperCase();
  if (content.substring(1) == "D3 94 BA 0F") //here is the UID of the RFID tag.
  {
    Serial.println("Authorized access");
    Serial.println();
    delay(500);
```

```
  {
    Serial.println("Authorized access");
    Serial.println();
    delay(500);
    digitalWrite(LED_G, HIGH);
    tone(BUZZER, 500);
    delay(300);
    noTone(BUZZER);
    myServo.write(180);
    delay(5000);
    myServo.write(0);
    digitalWrite(LED_G, LOW);
  }

 else   {
    Serial.println(" Access denied");
    digitalWrite(LED_R, HIGH);
    tone(BUZZER, 300);
    delay(1000);
    digitalWrite(LED_R, LOW);
    noTone(BUZZER);
  }
}
```




UNO

Performance Analysis

The RFID-based door locking system shows fast, reliable, and accurate performance during operation. The system responds immediately when an authorized RFID card is scanned, unlocking the door within a few seconds. Unauthorized cards are correctly detected, and access is denied with an alert indication, ensuring good security. The servo motor operates smoothly with accurate locking and unlocking angles. The system is stable for continuous use and consumes low power. When IoT features are enabled, real-time access data is transmitted with minimal delay. Overall, the system performs efficiently with high accuracy, quick response time, and dependable security control.

Applications:

RFID door lock systems have various real-life applications across different sectors, revolutionizing security measures. These applications include:

- **Residential Security:** RFID door locks provide enhanced security for homes, allowing residents to unlock doors conveniently using RFID tags like key cards.
- **Commercial Buildings:** In commercial settings, RFID door locks offer secure access control, enabling employees to enter designated areas with ease while maintaining high levels of security.
- **Industrial Facilities:** RFID technology is utilized in industrial environments to control access to sensitive areas, ensuring only authorized personnel can enter restricted zones.
- **Educational Institutions**: RFID door locks are implemented in schools and universities to enhance security measures, restricting access to specific areas and ensuring the safety of students and staff.
- **Hospitality Industry:** Hotels and resorts utilize RFID door lock systems for guest room entry, providing a secure and convenient way for guests to access their rooms while allowing hotel staff to monitor and manage access efficiently.
- **Healthcare Facilities**: RFID technology is employed in healthcare settings to control access to restricted areas like laboratories, pharmacies, and patient rooms, ensuring only authorized personnel can enter these spaces.
- **Government Buildings:** Government facilities use RFID door lock systems to regulate access to sensitive areas, enhancing security protocols and preventing unauthorized entry.

Advantages & Limitations

Advantages

- Provides high security using unique RFID authentication.
- Fast and contactless access, making it user-friendly.
- Low power consumption and cost-effective design.
- Reduces the risk of key loss or duplication.
- Can be easily upgraded with IoT features for remote monitoring.
- Suitable for homes, offices, labs, and restricted areas.

Limitations

- If an RFID card is lost or stolen, unauthorized access is possible.
- Limited security compared to biometric systems.
- Depends on a stable power supply for proper functioning.
- Basic Arduino systems have limited memory and processing power.
- Wi-Fi-based IoT features depend on internet availability.

Conclusion & Future Scope

Conclusion

The RFID-based door locking system successfully demonstrates a secure, fast, and automated access control mechanism using Arduino. It eliminates the drawbacks of traditional key-based systems by providing contactless authentication and automatic door operation. The system is reliable, cost-effective, and suitable for real-world security applications in homes, offices, and institutions.

Future Scope

In the future, this system can be enhanced by integrating biometric authentication such as fingerprint or face recognition for higher security. A mobile application can be developed for remote door control and monitoring. Cloud-based data storage, visitor logs, and AI-based access prediction can also be implemented to make the system smarter and more secure.

References:

- https://youtu.be/wHEwZ1uJExM?si=3PI97H_ItcaxaZNj
- https://youtu.be/GOO84CGBPz8?si=JA44_n-eLkkhnZXg
- https://srituhobby.com/how-to-make-a-rfid-door-lock-with-arduino/